

 Marwadi University Marwadi Chandarana Group	Marwadi University Faculty of Engineering & Technology Department of Information and Communication Technology	
Subject: Digital Signal and Image Processing(01CT1513)	Aim: Enhancement of Degraded Images Using Gray-Level Transformation Techniques	
Open Ended Problem: 07	Date: 14/08/2026	Enrollment No: 92400133147

Aim

To enhance five different degraded grayscale images using suitable gray-level transformation techniques and analyse the improvement in visual quality by comparing the original and enhanced images. The experiment focuses on selecting appropriate intensity transformations according to the degradation present in each image and studying their effectiveness and limitations. This directly addresses the assignment requirement to enhance each of the five degraded images and document the observations for every enhancement step.

OBJECTIVES

1. To understand the concept of gray-level transformations in digital image processing.
2. To load and convert degraded images into grayscale format.
3. To analyze the intensity distribution of each degraded image.
4. To select suitable gray-level transformation techniques for different types of degradation.
5. To implement image negative transformation.
6. To implement logarithmic transformation.
7. To implement power-law/gamma transformation.
8. To implement contrast stretching.
9. To implement piecewise-linear gray-level transformation where appropriate.
10. To compare the original and enhanced images.
11. To observe how different transformations affect brightness and contrast.
12. To determine which transformation gives the best visual improvement for each degraded image.
13. To understand the limitations of gray-level transformations.
14. To implement the complete experiment using Python, NumPy, OpenCV, and Matplotlib.

INTRODUCTION

Digital images may suffer from poor contrast, insufficient brightness, excessive darkness, or limited intensity variation. Such degradation can make important features difficult to observe.

Gray-level transformation is one of the fundamental techniques used in Digital Image Processing to modify the intensity values of pixels in an image.

A gray-level transformation can generally be represented as:

$$s = T(r)$$

where:

- r = input pixel intensity
- s = output pixel intensity
- T = transformation function

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For an 8-bit grayscale image:

$$0 \leq r \leq 255$$

and normally:

$$0 \leq s \leq 255$$

The purpose of the transformation is to map the original intensity values into a more useful range so that hidden details become more visible.

THEORY

A grayscale image contains only intensity information rather than separate RGB color components.

For an 8-bit grayscale image:

0 = black

255 = white

Intermediate values represent different shades of gray.

GRAY-LEVEL TRANSFORMATION TECHNIQUES

For this experiment, the techniques can be selected according to the visible degradation of each image.

1) Image Negative:

The negative transformation is:

$$s = L - 1 - r$$

For an 8-bit image:

$$s = 255 - r$$

Dark pixels become bright and bright pixels become dark.

Suitable for Images where important details are more visible in the negative representation.

2) Logarithmic Transformation

The logarithmic transformation is:

$$s = c \cdot \log(1+r)$$

where:

$$c = 255 / \log(1+r_{\max})$$

It expands low-intensity values and compresses high-intensity values.

Suitable for Images containing important details in dark regions.

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3) Power-Law / Gamma Transformation

The transformation is:

$$s = c.r^{\gamma}$$

A normalized implementation is:

$$s = 255 \left(r / 255 \right)^{\gamma}$$

For $\gamma < 1$: dark regions are brightened.

For $\gamma > 1$: the image becomes darker.

Suitable for Dark or underexposed images.

4) Contrast Stretching

Contrast stretching expands the range of pixel intensities.

A simple transformation is:

$$s = (r - r_{min} / r_{max} - r_{min}) * 255$$

This increases the separation between dark and bright regions.

Suitable for Low-contrast images.

5) Piecewise Linear Transformation

Different intensity ranges can be mapped independently.

For example:

$$s = \begin{cases} 0, & r < r_1 \\ \frac{r-r_1}{r_2-r_1}(s_2 - s_1) + s_1, & r_1 \leq r \leq r_2 \\ 255, & r > r_2 \end{cases}$$

This allows selected intensity ranges to be enhanced more strongly.

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GENERAL ALGORITHM

1. Start the program.
2. Import OpenCV, NumPy, and Matplotlib.
3. Load the five degraded images.
4. Verify that each image is loaded successfully.
5. Convert each image into grayscale.
6. Analyze the visual characteristics of each image.
7. Select suitable gray-level transformations.
8. Apply the selected transformations.
9. Clip output values to the range 0–255.
10. Convert the results to uint8.
11. Display the original and enhanced images.
12. Save the enhanced images.
13. Compare the visual quality.
14. Record observations.
15. Identify the most effective transformation for each image.
16. Discuss the limitations.
17. Draw the final conclusion.

Code:

Importing Required Libraries:

```

import cv2
import numpy as np
import matplotlib.pyplot as plt
import os

print("Libraries imported successfully")

```

[1] ✓ 0.6s

Output:

```

[1] ✓ 0.6s

... Libraries imported successfully

```

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Defining Image Paths

Code:

```

input_folder = r"C:\Users\devah\Documents\ICT MU\Sem 5\DSIP\Lab\Images"
output_folder = r"C:\Users\devah\Documents\ICT MU\Sem 5\DSIP\Lab\Enhanced Images"
os.makedirs(output_folder, exist_ok=True)
print("Folders ready")

```

[2] ✓ 0.0s

Output:

```

[2] ✓ 0.0s
... Folders ready

```

Loading Five Images

Code:

```

image1 = cv2.imread(os.path.join(input_folder, "1.png"))
image2 = cv2.imread(os.path.join(input_folder, "2.png"))
image3 = cv2.imread(os.path.join(input_folder, "3.png"))
image4 = cv2.imread(os.path.join(input_folder, "4.png"))
image5 = cv2.imread(os.path.join(input_folder, "5.png"))

images = [image1, image2, image3, image4, image5]

for i, img in enumerate(images, 1):
    if img is None:
        print("Image", i, "not found")
    else:
        print("Image", i, "loaded successfully")

```

[5] ✓ 0.0s

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Output:

```
[5] ✓ 0.0s
... Image 1 loaded successfully
Image 2 loaded successfully
Image 3 loaded successfully
Image 4 loaded successfully
Image 5 loaded successfully
```

Conversion of Degraded Images to Grayscale

Code:

```

▶
gray1 = cv2.cvtColor(image1, cv2.COLOR_BGR2GRAY)
gray2 = cv2.cvtColor(image2, cv2.COLOR_BGR2GRAY)
gray3 = cv2.cvtColor(image3, cv2.COLOR_BGR2GRAY)
gray4 = cv2.cvtColor(image4, cv2.COLOR_BGR2GRAY)
gray5 = cv2.cvtColor(image5, cv2.COLOR_BGR2GRAY)

gray_images = [gray1, gray2, gray3, gray4, gray5]

print("All images converted to grayscale")
[6] ✓ 0.0s

```

Output:

```
[6] ✓ 0.0s
... All images converted to grayscale
```

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Original Grayscale Images

Code:

```

plt.figure(figsize=(15, 8))

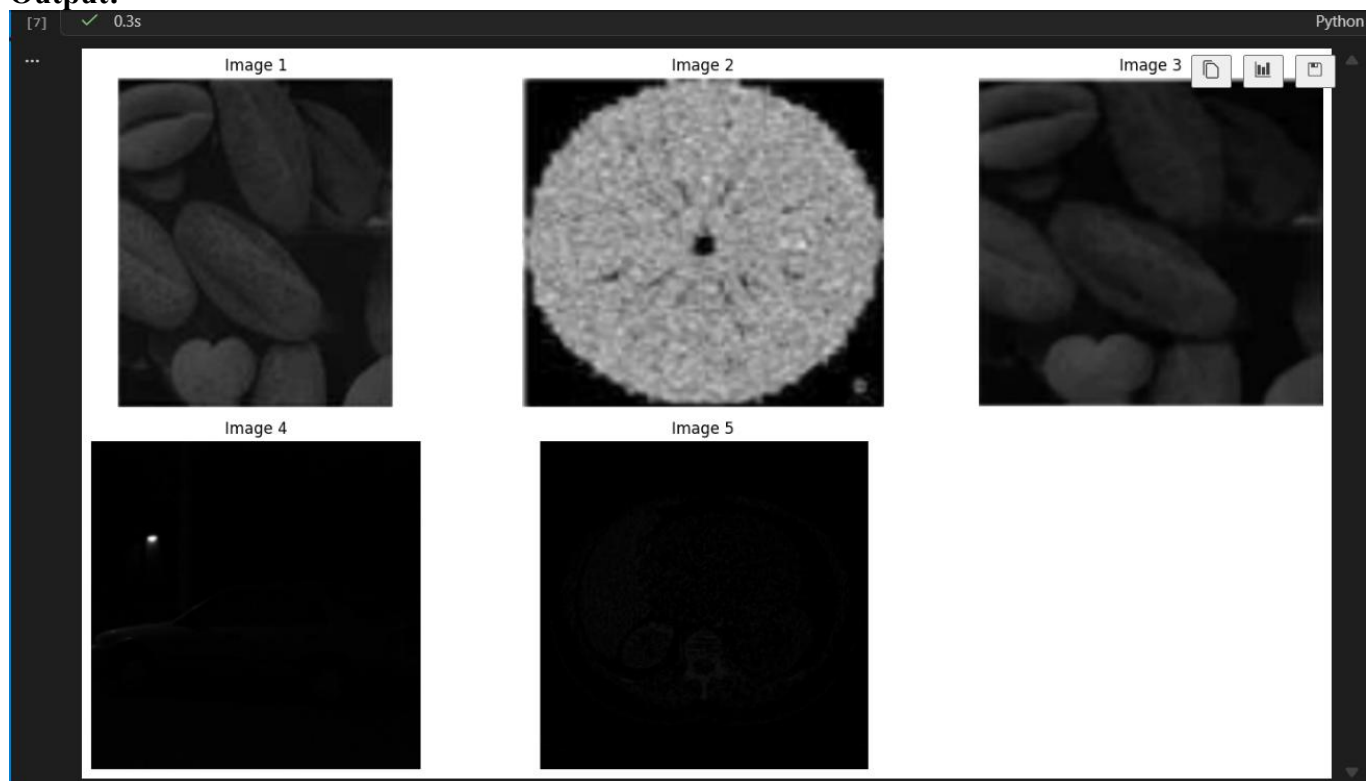
for i, img in enumerate(gray_images, 1):
    plt.subplot(2, 3, i)
    plt.imshow(img, cmap="gray")
    plt.title("Image " + str(i))
    plt.axis("off")

plt.tight_layout()
plt.show()

```

[7] ✓ 0.3s

Output:



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Image 1

Objective

To enhance the contrast of the first degraded image using suitable gray - level transformation techniques and improve the visibility and separation of objects from the background.

Theory

The first image has **low contrast**, where the gray-level values occupy a relatively narrow range. Because the difference between the dark and bright regions is small, individual objects are not clearly distinguishable.

A suitable transformation is **contrast stretching**.

The contrast-stretching equation is:

where:

- r = input gray level
- s = output gray level
- r_{min} = minimum input intensity
- r_{max} = maximum input intensity

This expands the existing gray-level range to approximately 0–255.

Additional methods

Gamma transformation:

$$s = 255(r / 255) ^ \gamma$$

For:

$$\Gamma < 1$$

dark regions are brightened.

Log transformation:

$$s = c. \log(1+r)$$

which expands lower gray-level values.

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Suitable Methods

1. Original grayscale image
2. Contrast stretching
3. Gamma transformation
4. Logarithmic transformation

Primary method: Contrast stretching.

Algorithm

1. Read the first degraded image.
2. Convert it to grayscale.
3. Find the minimum and maximum pixel intensities.
4. Apply contrast stretching.
5. Apply gamma transformation with .
6. Apply logarithmic transformation.
7. Normalize the output to 0–255.
8. Display the original and enhanced images.
9. Compare the results.
10. Select the most effective transformation.

Display Image 1

Code:

```

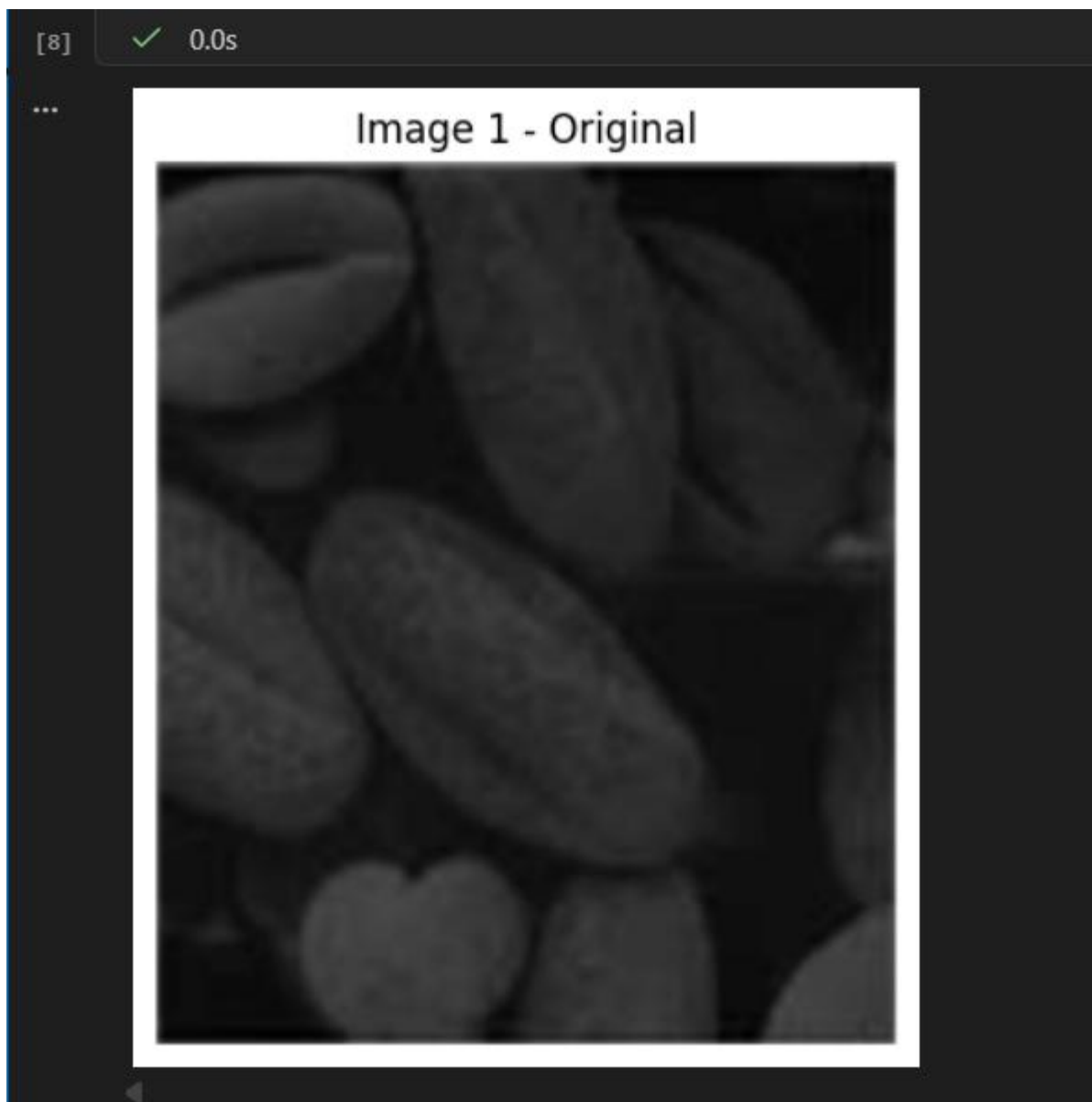
plt.figure(figsize=(6, 5))
plt.imshow(gray1, cmap="gray")
plt.title("Image 1 - Original")
plt.axis("off")
plt.show()

```

[8] ✓ 0.0s

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Contrast Stretching

Code:

```

r_min = np.min(gray1)
r_max = np.max(gray1)

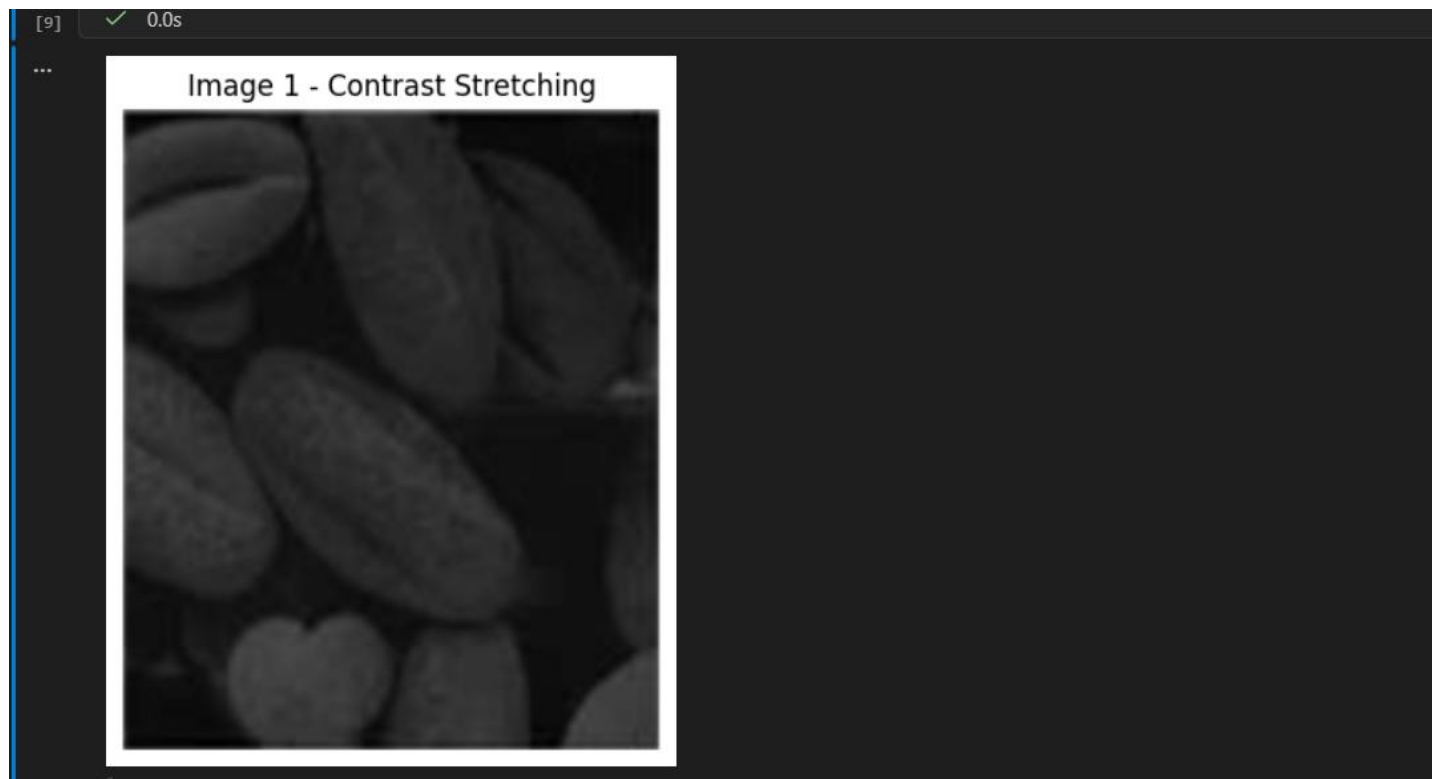
contrast1 = ((gray1 - r_min) / (r_max - r_min) * 255).astype(np.uint8)

cv2.imwrite(
    os.path.join(output_folder, "image1_contrast_stretching.png"),
    contrast1
)

plt.figure(figsize=(6, 5))
plt.imshow(contrast1, cmap="gray")
plt.title("Image 1 - Contrast Stretching")
plt.axis("off")
plt.show()

```

Output:



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Gamma Transformation

Code:

```

gamma = 0.6

gamma1 = 255 * (gray1.astype(np.float32) / 255) ** gamma
gamma1 = np.clip(gamma1, 0, 255).astype(np.uint8)

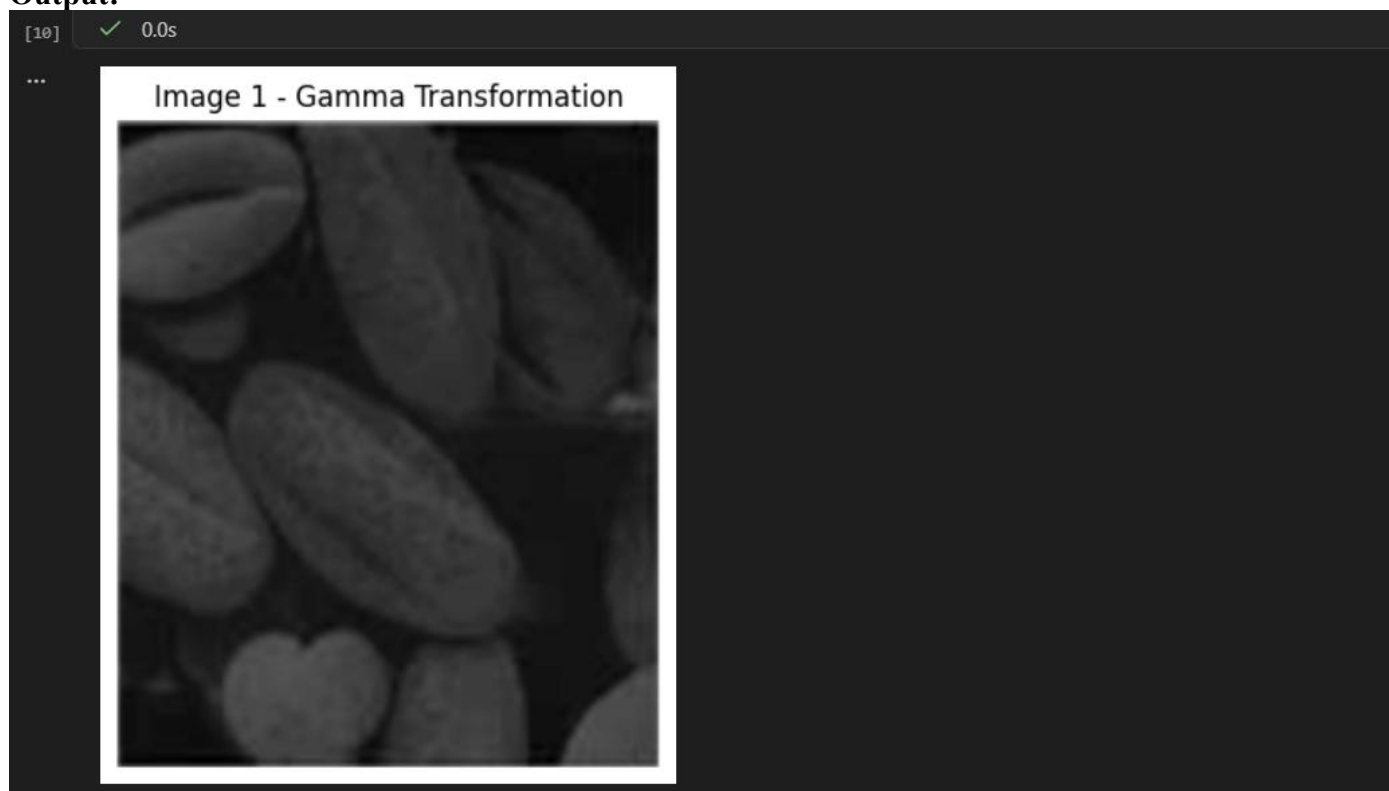
cv2.imwrite(
    os.path.join(output_folder, "image1_gamma.png"),
    gamma1
)

plt.figure(figsize=(6, 5))
plt.imshow(gamma1, cmap="gray")
plt.title("Image 1 - Gamma Transformation")
plt.axis("off")
plt.show()

```

[10] ✓ 0.0s

Output:



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Log Transformation

Code:

```

c = 255 / np.log(1 + np.max(gray1))

log1 = c * np.log(1 + gray1.astype(np.float32))
log1 = np.clip(log1, 0, 255).astype(np.uint8)

cv2.imwrite(
    os.path.join(output_folder, "image1_log.png"),
    log1
)

plt.figure(figsize=(6, 5))
plt.imshow(log1, cmap="gray")
plt.title("Image 1 - Log Transformation")
plt.axis("off")
plt.show()

```

[11] ✓ 0.0s Python

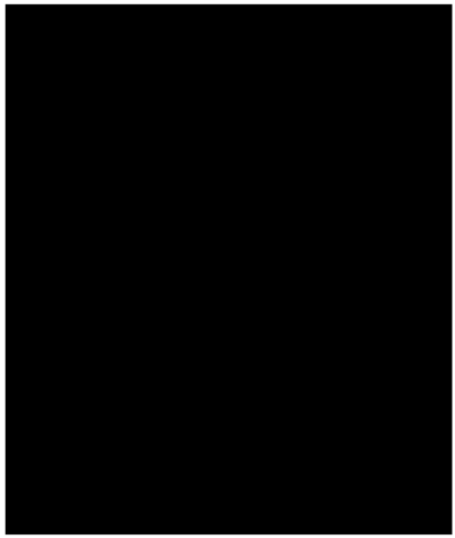
Output:

```

[11] ✓ 0.0s Python
... C:\Users\devah\AppData\Local\Temp\ipykernel_26292\3680458346.py:1: RuntimeWarning: overflow encountered in scalar add
c = 255 / np.log(1 + np.max(gray1))
C:\Users\devah\AppData\Local\Temp\ipykernel_26292\3680458346.py:1: RuntimeWarning: divide by zero encountered in log
c = 255 / np.log(1 + np.max(gray1))
...

```

Image 1 - Log Transformation



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Image 1 Comparison

Code:

```
plt.figure(figsize=(12, 8))

plt.subplot(2, 2, 1)
plt.imshow(gray1, cmap="gray")
plt.title("Original")
plt.axis("off")

plt.subplot(2, 2, 2)
plt.imshow(contrast1, cmap="gray")
plt.title("Contrast Stretching")
plt.axis("off")

plt.subplot(2, 2, 3)
plt.imshow(gamma1, cmap="gray")
plt.title("Gamma Transformation")
plt.axis("off")

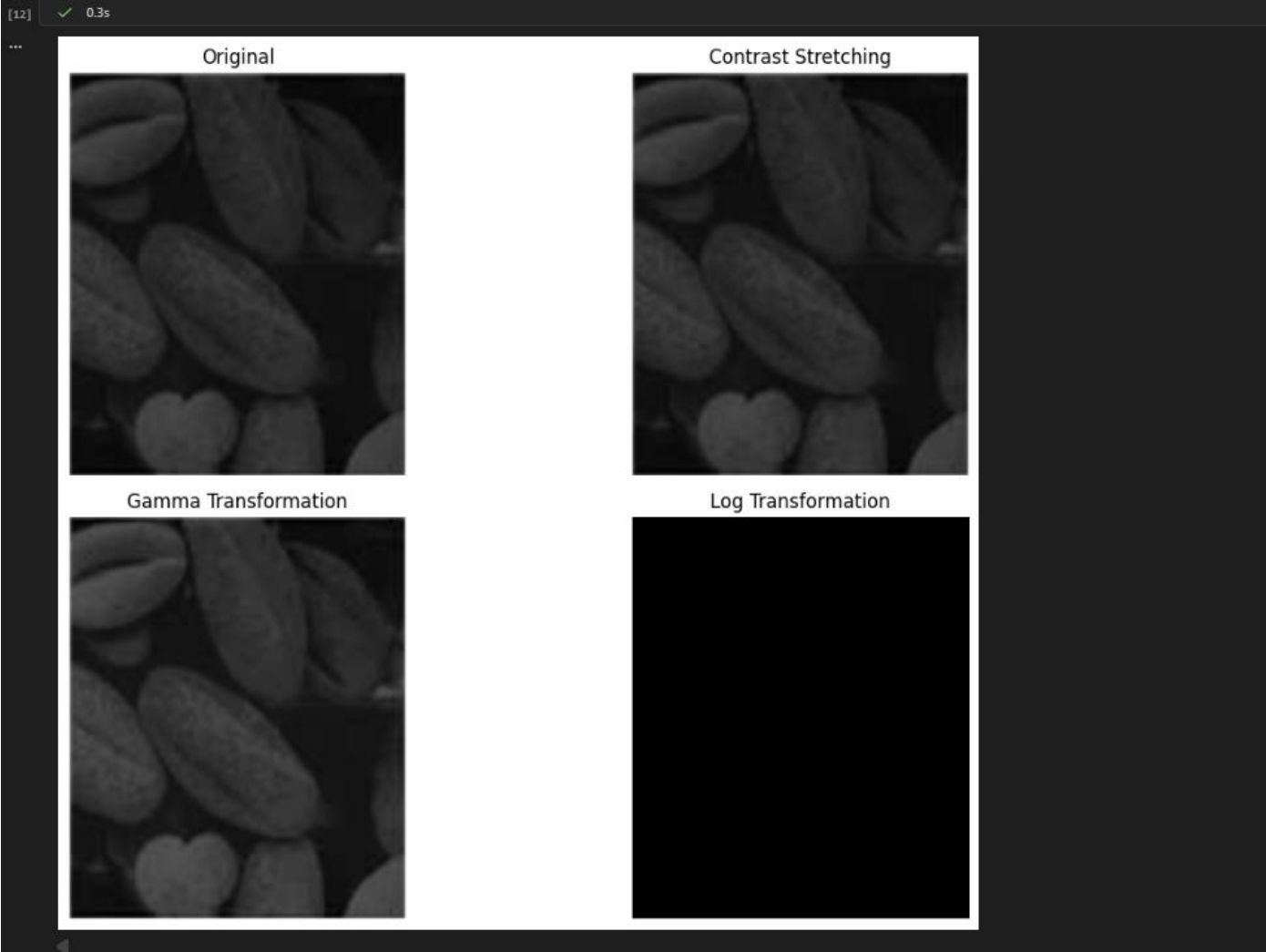
plt.subplot(2, 2, 4)
plt.imshow(log1, cmap="gray")
plt.title("Log Transformation")
plt.axis("off")

plt.tight_layout()
plt.show()
```

[12] ✓ 0.3s

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Output:



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Observation – Image 1

- The original image has poor contrast.
- Contrast stretching increases the difference between dark and bright regions.
- Objects become more distinguishable after contrast stretching.
- Gamma transformation with increases the brightness of darker regions.
- Log transformation also enhances low-intensity information.
- Contrast stretching provides better overall contrast improvement for this image.

Conclusion – Image 1

The first image was successfully enhanced using gray-level transformations. **Contrast stretching is the most suitable technique** because the major problem is limited contrast. Gamma and logarithmic transformations provide additional enhancement, particularly in darker regions.

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IMAGE 2

Objective

To enhance the visibility of structures in the second degraded image while studying the effect of gray-level transformations on a noisy medical image.

Theory

The second image is a **CT-like medical image with significant granular/noisy appearance**. The image contains useful structural information, but the noise makes the structures less clear. Gray-level transformations can improve the visibility of existing structures, but they **do not perform true noise removal**.

Log Transformation

$$s=c\log(1+r)$$

This expands lower intensity values.

Gamma Transformation

$$s=cr^{\gamma}$$

With:

$$\gamma < 1$$

dark areas become brighter.

Contrast Stretching

$$s = (r - r_{min} / r_{max} - r_{min}) * 255$$

This increases the intensity range.

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Suitable Methods

1. Original grayscale
2. Log transformation
3. Gamma transformation
4. Contrast stretching

Primary method: Contrast stretching.

Algorithm

1. Read the CT-like image.
2. Convert it to grayscale.
3. Analyze its intensity range.
4. Apply logarithmic transformation.
5. Apply gamma transformation.
6. Apply contrast stretching.
7. Display all results.
8. Compare structural visibility.
9. Observe the effect of noise.
10. Select the most useful enhancement.

Original Image:

Code:

```

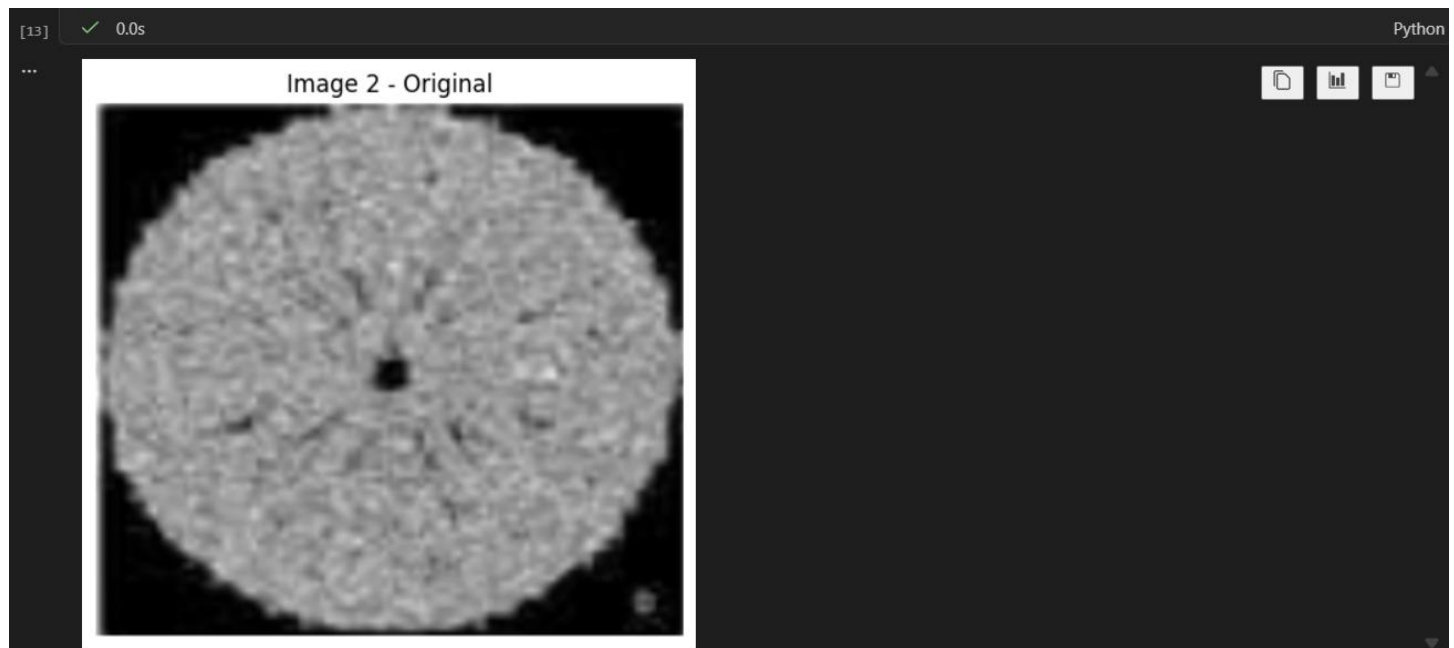
plt.figure(figsize=(6, 5))
plt.imshow(gray2, cmap="gray")
plt.title("Image 2 - Original")
plt.axis("off")
plt.show()

```

[13] ✓ 0.0s

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Output:



Log Transformation

Code:

```

c = 255 / np.log(1 + np.max(gray2))

log2 = c * np.log(1 + gray2.astype(np.float32))
log2 = np.clip(log2, 0, 255).astype(np.uint8)

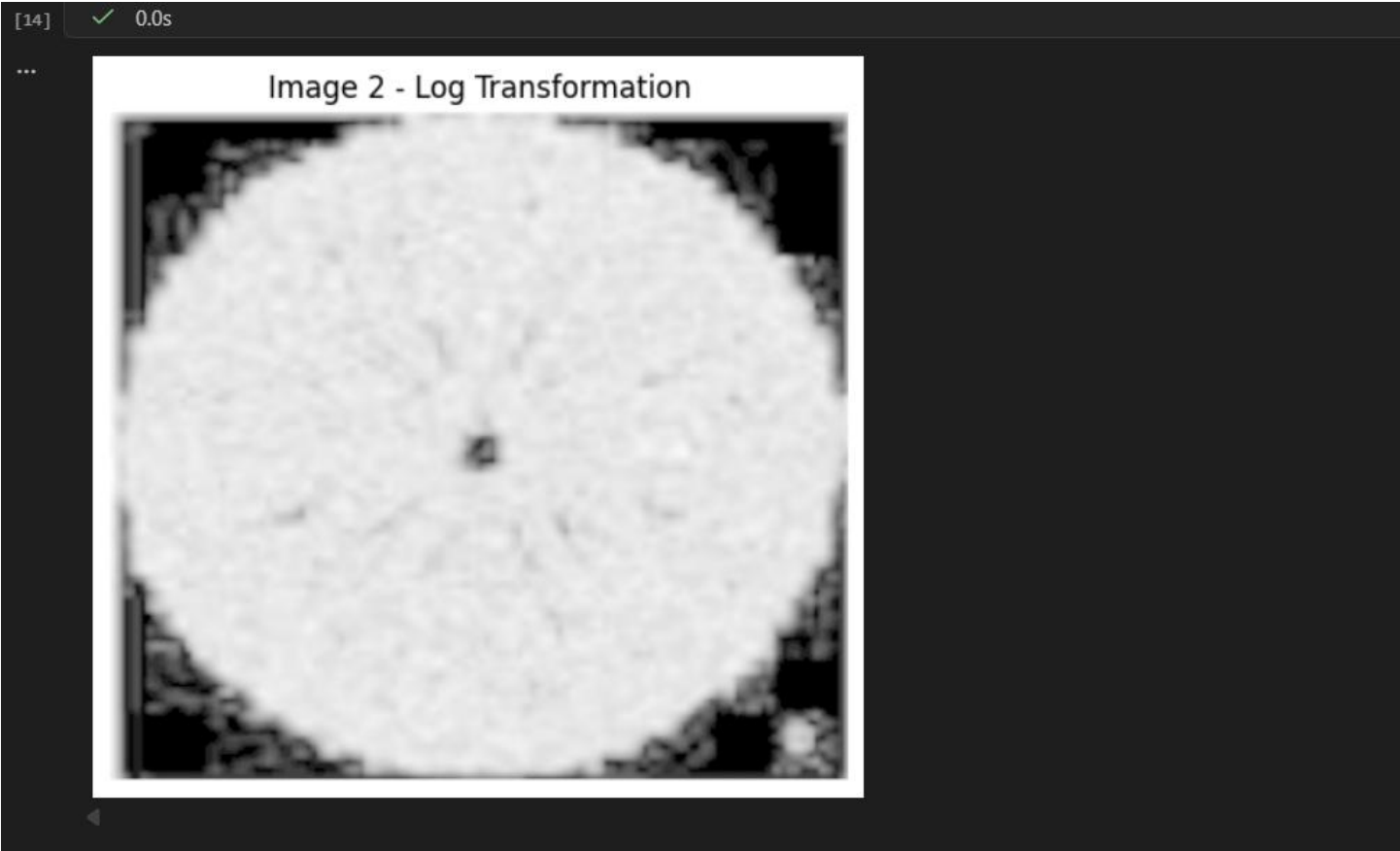
cv2.imwrite(
    os.path.join(output_folder, "image2_log.png"),
    log2
)

plt.figure(figsize=(6, 5))
plt.imshow(log2, cmap="gray")
plt.title("Image 2 - Log Transformation")
plt.axis("off")
plt.show()

```

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Gamma Transformation

Code:

```

gamma = 0.7

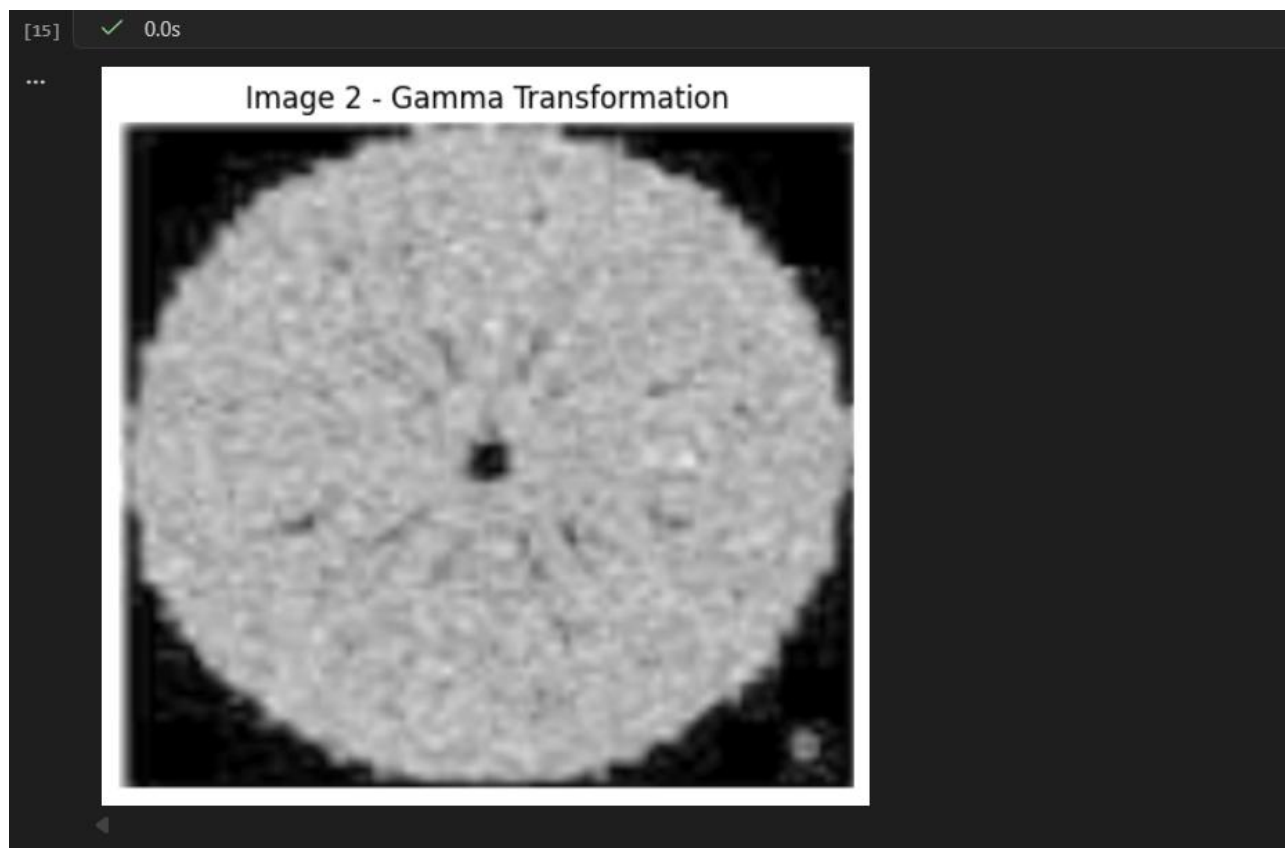
gamma2 = 255 * (gray2.astype(np.float32) / 255) ** gamma
gamma2 = np.clip(gamma2, 0, 255).astype(np.uint8)

cv2.imwrite(
    os.path.join(output_folder, "image2_gamma.png"),
    gamma2
)

plt.figure(figsize=(6, 5))
plt.imshow(gamma2, cmap="gray")
plt.title("Image 2 - Gamma Transformation")
plt.axis("off")
plt.show()

```

Output:



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Contrast Stretching

Code:

```

r_min = np.min(gray2)
r_max = np.max(gray2)

contrast2 = ((gray2 - r_min) / (r_max - r_min) * 255).astype(np.uint8)

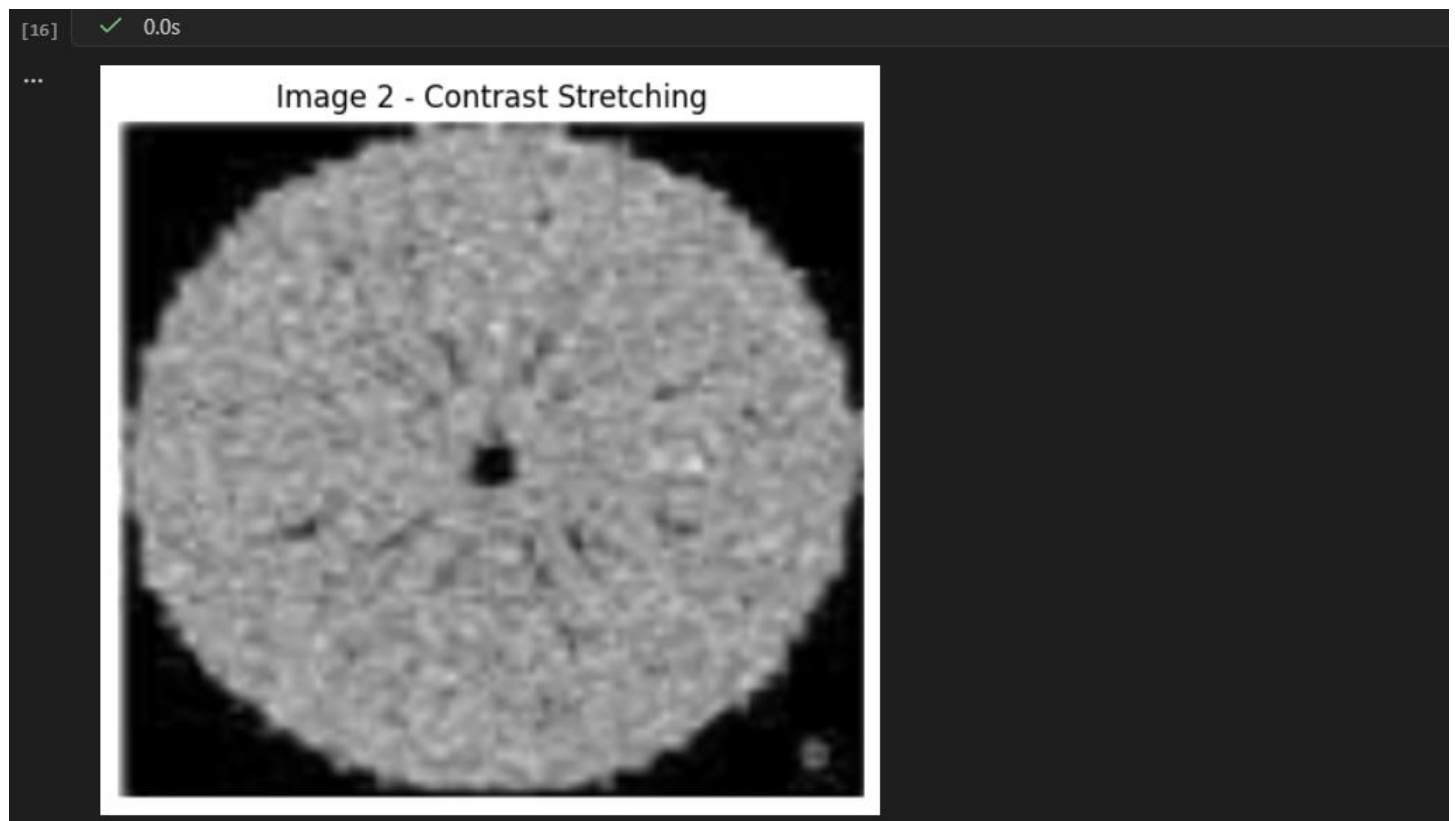
cv2.imwrite(
    os.path.join(output_folder, "image2_contrast.png"),
    contrast2
)

plt.figure(figsize=(6, 5))
plt.imshow(contrast2, cmap="gray")
plt.title("Image 2 - Contrast Stretching")
plt.axis("off")
plt.show()

```

[16] ✓ 0.0s Python

Output:



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Observation – Image 2

- The original image contains considerable granular variation.
- Log transformation enhances lower-intensity structures.
- Gamma transformation changes the brightness distribution.
- Contrast stretching improves the overall intensity range.
- Enhancement can make the structures more visible.
- However, the existing noise may also become more noticeable.

Conclusion – Image 2

Gray-level transformations improve the visibility of structures in the second image, but they **cannot effectively remove the granular noise**. This demonstrates that intensity enhancement and noise reduction are different processes. For better results, gray-level enhancement could be followed by suitable noise-reduction filtering.

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Image 2 Comparison

Code:

```

plt.figure(figsize=(12, 8))

plt.subplot(2, 2, 1)
plt.imshow(gray2, cmap="gray")
plt.title("Original Image 2")
plt.axis("off")

plt.subplot(2, 2, 2)
plt.imshow(contrast2, cmap="gray")
plt.title("Contrast Stretching")
plt.axis("off")

plt.subplot(2, 2, 3)
plt.imshow(gamma2, cmap="gray")
plt.title("Gamma Transformation")
plt.axis("off")

plt.subplot(2, 2, 4)
plt.imshow(log2, cmap="gray")
plt.title("Log Transformation")
plt.axis("off")

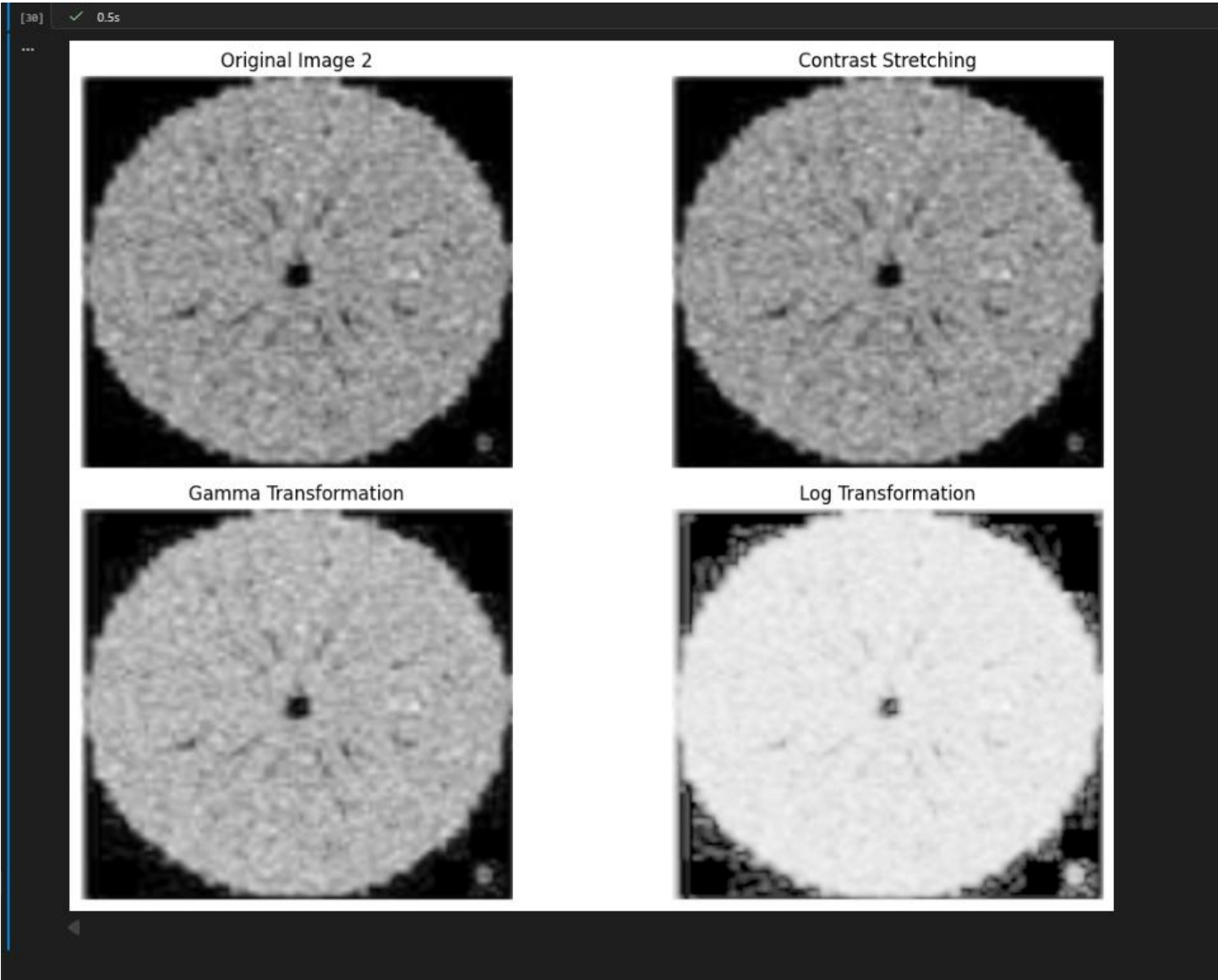
plt.tight_layout()
plt.show()

```

[30] ✓ 0.5s

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IMAGE 3

Objective

To enhance the third degraded image, which contains poorly illuminated objects, by increasing the visibility of low-intensity regions.

Theory

The third image is considerably dark. Most of the useful information is concentrated toward the lower end of the gray-level range.

For such an image, gamma transformation with $\gamma < 1$ is particularly useful.

$$s = 255(r / 255) ^ \gamma$$

For:

$$\gamma < 1$$

the output intensity becomes greater than the input for most non-zero values.

Log transformation can also be used:

$$s = clog(1+r)$$

Both methods expand the darker gray levels.

Suitable Methods

1. Original grayscale
2. Gamma transformation
3. Log transformation
4. Contrast stretching

Primary method: Gamma transformation.

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Algorithm

1. Read Image 3.
2. Convert it to grayscale.
3. Select a gamma value below 1.
4. Apply gamma transformation.
5. Apply logarithmic transformation.
6. Apply contrast stretching.
7. Display the outputs.
8. Compare the visibility of dark objects.
9. Select the most effective result.

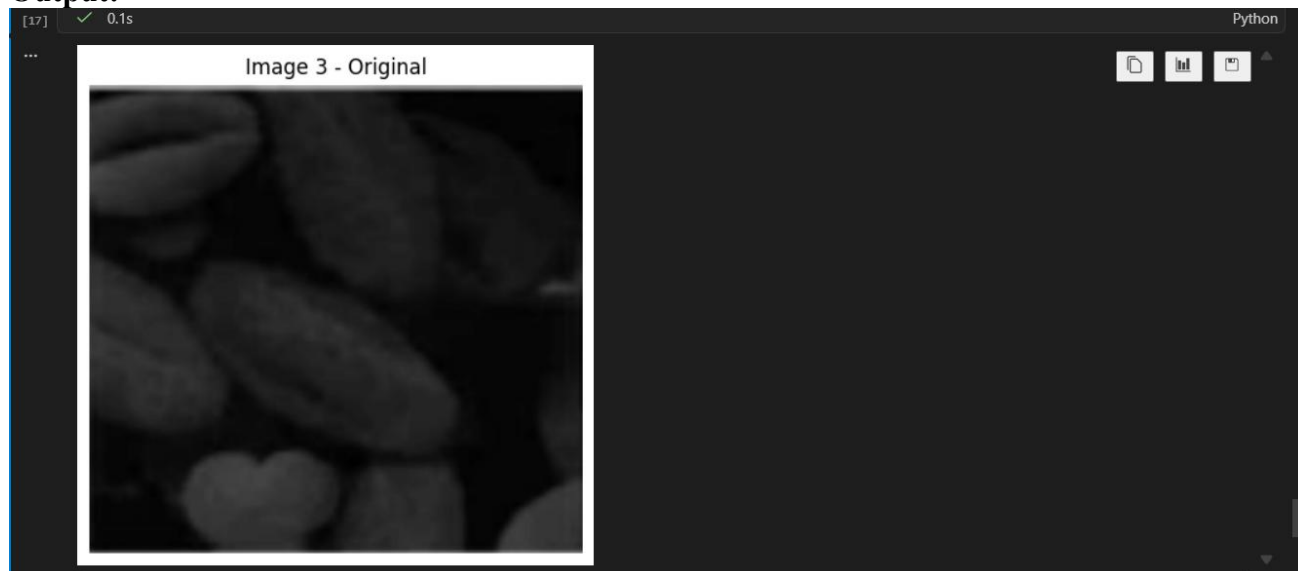
Original Image

Code:

```
plt.figure(figsize=(6, 5))
plt.imshow(gray3, cmap="gray")
plt.title("Image 3 - Original")
plt.axis("off")
plt.show()
```

[17] ✓ 0.1s

Output:



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Gamma Transformation

Code:

```

gamma = 0.4

gamma3 = 255 * (gray3.astype(np.float32) / 255) ** gamma
gamma3 = np.clip(gamma3, 0, 255).astype(np.uint8)

cv2.imwrite(
    os.path.join(output_folder, "image3_gamma.png"),
    gamma3
)

plt.figure(figsize=(6, 5))
plt.imshow(gamma3, cmap="gray")
plt.title("Image 3 - Gamma Transformation")
plt.axis("off")
plt.show()

```

[18] ✓ 0.1s

Output:



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Log Transformation

Code:

```

c = 255 / np.log(1 + np.max(gray3))

log3 = c * np.log(1 + gray3.astype(np.float32))
log3 = np.clip(log3, 0, 255).astype(np.uint8)

cv2.imwrite(
    os.path.join(output_folder, "image3_log.png"),
    log3
)

plt.figure(figsize=(6, 5))
plt.imshow(log3, cmap="gray")
plt.title("Image 3 - Log Transformation")
plt.axis("off")
plt.show()

```

[19] ✓ 0.1s

Output:



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Contrast Stretching

Code:

```

r_min = np.min(gray3)
r_max = np.max(gray3)

contrast3 = ((gray3 - r_min) / (r_max - r_min) * 255).astype(np.uint8)

cv2.imwrite(
    os.path.join(output_folder, "image3_contrast.png"),
    contrast3
)

plt.figure(figsize=(6, 5))
plt.imshow(contrast3, cmap="gray")
plt.title("Image 3 - Contrast Stretching")
plt.axis("off")
plt.show()

```

[20] ✓ 0.1s

Output:



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Image 3 Comparison

Code:

```

plt.figure(figsize=(12, 8))

plt.subplot(2, 2, 1)
plt.imshow(gray3, cmap="gray")
plt.title("Original Image 3")
plt.axis("off")

plt.subplot(2, 2, 2)
plt.imshow(contrast3, cmap="gray")
plt.title("Contrast Stretching")
plt.axis("off")

plt.subplot(2, 2, 3)
plt.imshow(gamma3, cmap="gray")
plt.title("Gamma Transformation")
plt.axis("off")

plt.subplot(2, 2, 4)
plt.imshow(log3, cmap="gray")
plt.title("Log Transformation")
plt.axis("off")

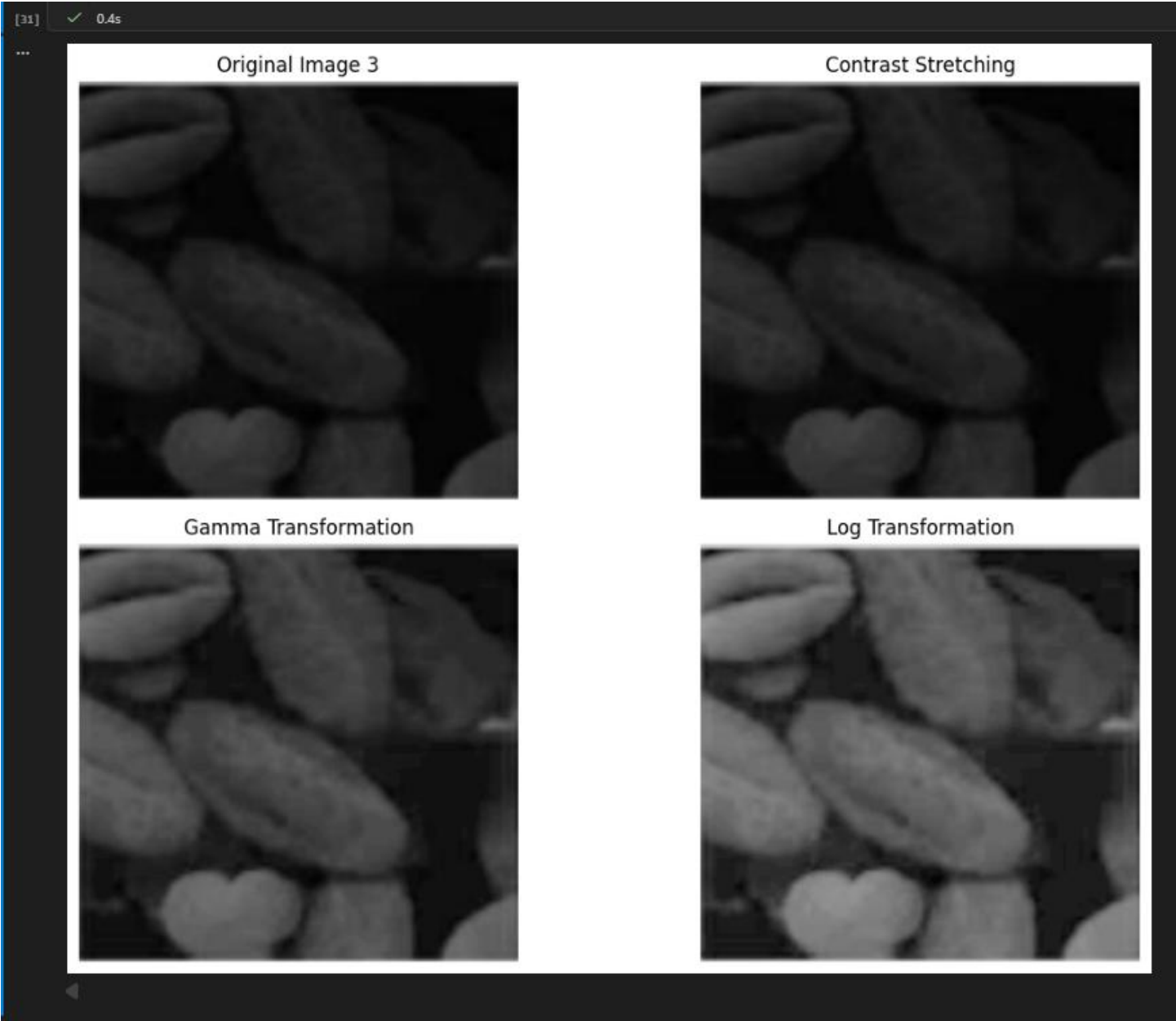
plt.tight_layout()
plt.show()

```

[31] ✓ 0.4s

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Output:



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Observation

- The original image is dark and objects have poor visibility.
- Gamma transformation with significantly increases the brightness of dark regions.
- Log transformation also expands low-intensity values.
- Contrast stretching increases the overall dynamic range.
- More details become visible after enhancement.

Conclusion

Gamma transformation with is particularly effective for the third image because the major degradation is insufficient brightness. Log transformation also provides useful low-intensity enhancement.

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IMAGE 4

Objective

To improve the visibility of extremely dark regions in the fourth degraded image using suitable gray-level transformations.

Theory

The fourth image contains a very large dark region with only a small bright source and faint structures. The useful information is concentrated in a small portion of the intensity range. For this type of image, transformations that expand low gray levels are appropriate.

Gamma transformation

$$s = 255(r / 255) ^ \gamma$$

Using a small value such as:

$$\gamma = 0.3$$

can strongly brighten dark areas.

Log transformation

$$s = c.\log(1+r)$$

also expands low-intensity values.

Suitable Methods

1. Original grayscale
2. Gamma transformation
3. Log transformation
4. Contrast stretching

Primary method: Gamma transformation.

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Algorithm

1. Read Image 4.
2. Convert it into grayscale.
3. Apply gamma transformation with $\gamma < 1$
4. Apply logarithmic transformation.
5. Apply contrast stretching.
6. Display all outputs.
7. Compare the visibility of faint structures.
8. Analyse possible over-enhancement.
9. Select the best result.

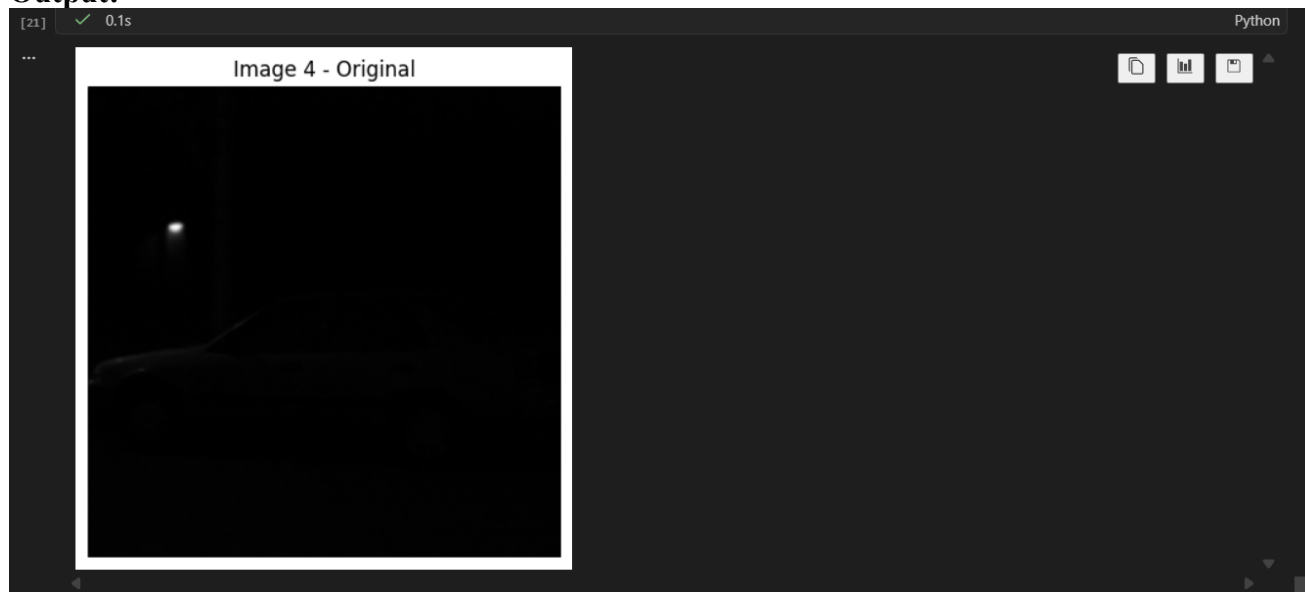
Original Image

Code:

```
plt.figure(figsize=(6, 5))
plt.imshow(gray4, cmap="gray")
plt.title("Image 4 - Original")
plt.axis("off")
plt.show()
```

[21] ✓ 0.1s

Output:



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Log Transformation

Code:

```

c = 255 / np.log(1 + np.max(gray4))

log4 = c * np.log(1 + gray4.astype(np.float32))
log4 = np.clip(log4, 0, 255).astype(np.uint8)

cv2.imwrite(
    os.path.join(output_folder, "image4_log.png"),
    log4
)

plt.figure(figsize=(6, 5))
plt.imshow(log4, cmap="gray")
plt.title("Image 4 - Log Transformation")
plt.axis("off")
plt.show()

```

Output:



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Gamma Transformation

Code:

```

gamma = 0.3

gamma4 = 255 * (gray4.astype(np.float32) / 255) ** gamma
gamma4 = np.clip(gamma4, 0, 255).astype(np.uint8)

cv2.imwrite(
    os.path.join(output_folder, "image4_gamma.png"),
    gamma4
)

plt.figure(figsize=(6, 5))
plt.imshow(gamma4, cmap="gray")
plt.title("Image 4 - Gamma Transformation")
plt.axis("off")
plt.show()

```

[23] ✓ 0.1s

Output:



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Contrast Stretching

Code:

```

r_min = np.min(gray4)
r_max = np.max(gray4)

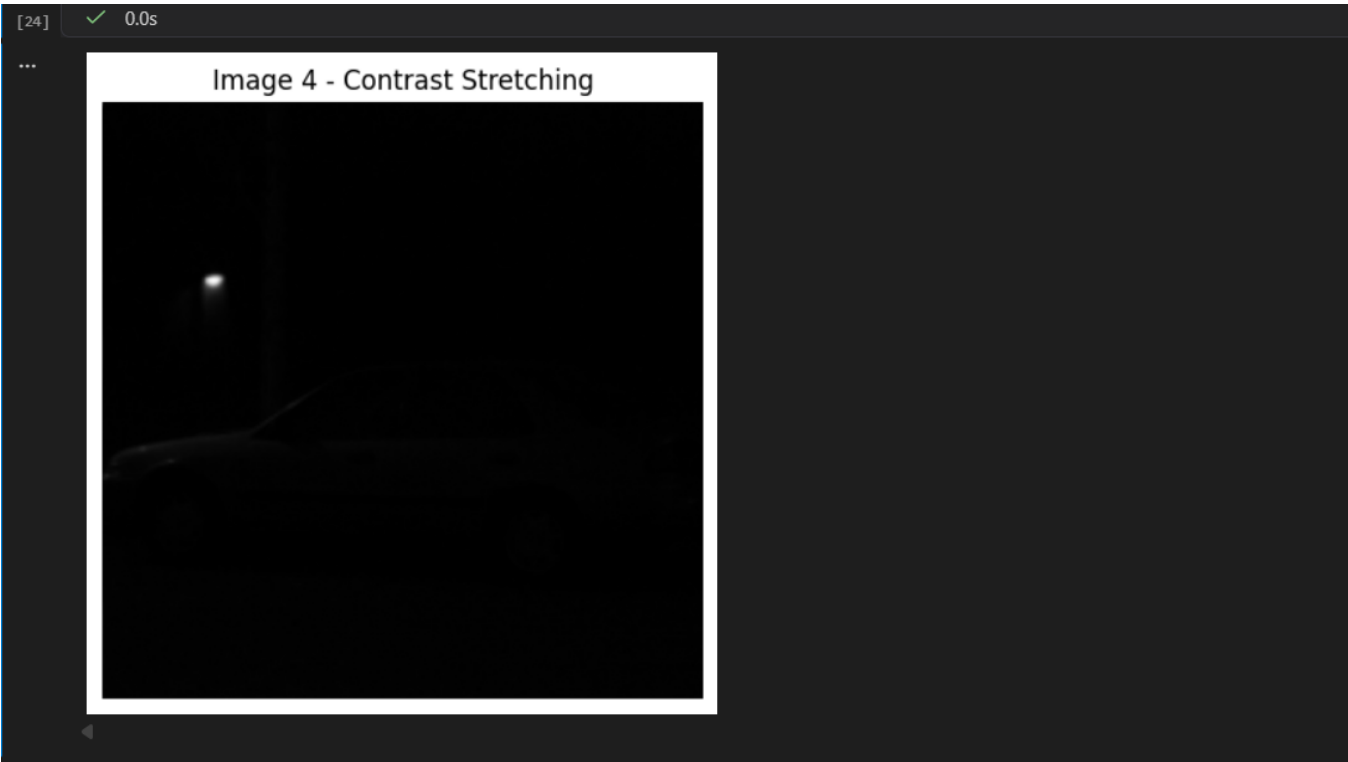
contrast4 = ((gray4 - r_min) / (r_max - r_min) * 255).astype(np.uint8)

cv2.imwrite(
    os.path.join(output_folder, "image4_contrast.png"),
    contrast4
)

plt.figure(figsize=(6, 5))
plt.imshow(contrast4, cmap="gray")
plt.title("Image 4 - Contrast Stretching")
plt.axis("off")
plt.show()

```

Output:



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Image 4 Comparison

Code:

```

plt.figure(figsize=(12, 8))

plt.subplot(2, 2, 1)
plt.imshow(gray4, cmap="gray")
plt.title("Original Image 4")
plt.axis("off")

plt.subplot(2, 2, 2)
plt.imshow(contrast4, cmap="gray")
plt.title("Contrast Stretching")
plt.axis("off")

plt.subplot(2, 2, 3)
plt.imshow(gamma4, cmap="gray")
plt.title("Gamma Transformation")
plt.axis("off")

plt.subplot(2, 2, 4)
plt.imshow(log4, cmap="gray")
plt.title("Log Transformation")
plt.axis("off")

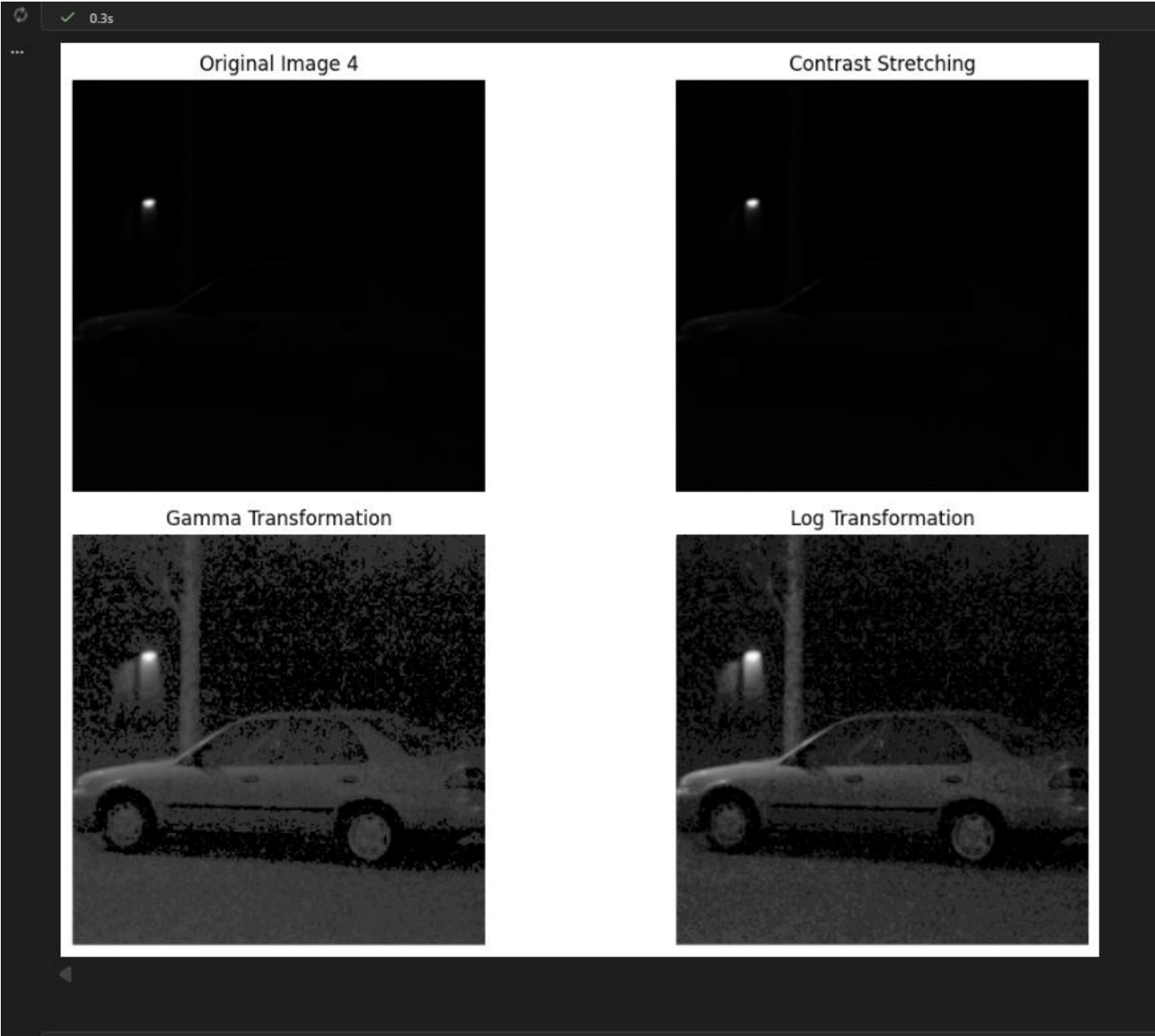
plt.tight_layout()
plt.show()

```

✓ 0.3s

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Output:



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Observation – Image 4

- The original image is extremely dark.
- The bright source is visible, but most surrounding information is difficult to observe.
- Gamma transformation substantially brightens dark regions.
- Log transformation also reveals low-intensity information.
- Contrast stretching expands the available intensity range.
- Strong enhancement may reveal noise and artifacts.

Conclusion – Image 4

Gamma and logarithmic transformations are suitable for the fourth image because they expand low-intensity information. However, if the original image contains very little captured information, enhancement cannot recover details that were never recorded.

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Image 5

Objective

To enhance the visibility of faint anatomical structures in the fifth degraded medical image using appropriate gray-level transformations.

Theory

The fifth image is extremely dark and contains faint medical/anatomical structures. Such an image requires enhancement of the lower gray-level range.

Gamma transformation with:

$$\gamma < 1$$

is useful because it brightens dark pixels.

The logarithmic transformation:

$$s = c \log(1+r)$$

also expands low-intensity values.

Contrast stretching can further improve the separation between different gray-level regions.

Suitable Methods

1. Original grayscale
2. Gamma transformation
3. Log transformation
4. Contrast stretching

Primary method: Gamma transformation.

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Algorithm

1. Load Image 5.
2. Convert it to grayscale.
3. Apply gamma transformation with .
4. Apply logarithmic transformation.
5. Apply contrast stretching.
6. Display the original and enhanced results.
7. Compare anatomical structure visibility.
8. Determine the most effective transformation.
9. Record limitations.

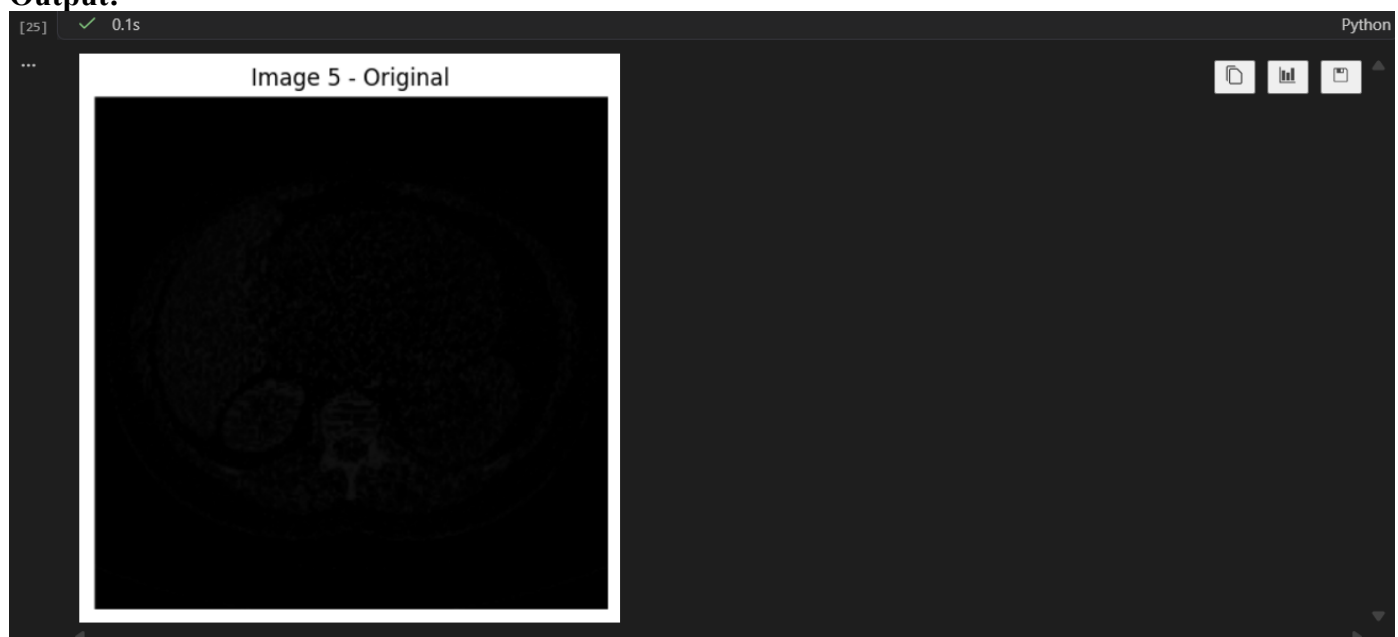
Original Image

Code:

```
plt.figure(figsize=(6, 5))
plt.imshow(gray5, cmap="gray")
plt.title("Image 5 - Original")
plt.axis("off")
plt.show()
```

[25] ✓ 0.1s

Output:



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Gamma Transformation

Code:

```

gamma = 0.4

gamma5 = 255 * (gray5.astype(np.float32) / 255) ** gamma
gamma5 = np.clip(gamma5, 0, 255).astype(np.uint8)

cv2.imwrite(
    os.path.join(output_folder, "image5_gamma.png"),
    gamma5
)

plt.figure(figsize=(6, 5))
plt.imshow(gamma5, cmap="gray")
plt.title("Image 5 - Gamma Transformation")
plt.axis("off")
plt.show()

```

Output:



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Log Transformation

Code:

```

c = 255 / np.log(1 + np.max(gray5))

log5 = c * np.log(1 + gray5.astype(np.float32))
log5 = np.clip(log5, 0, 255).astype(np.uint8)

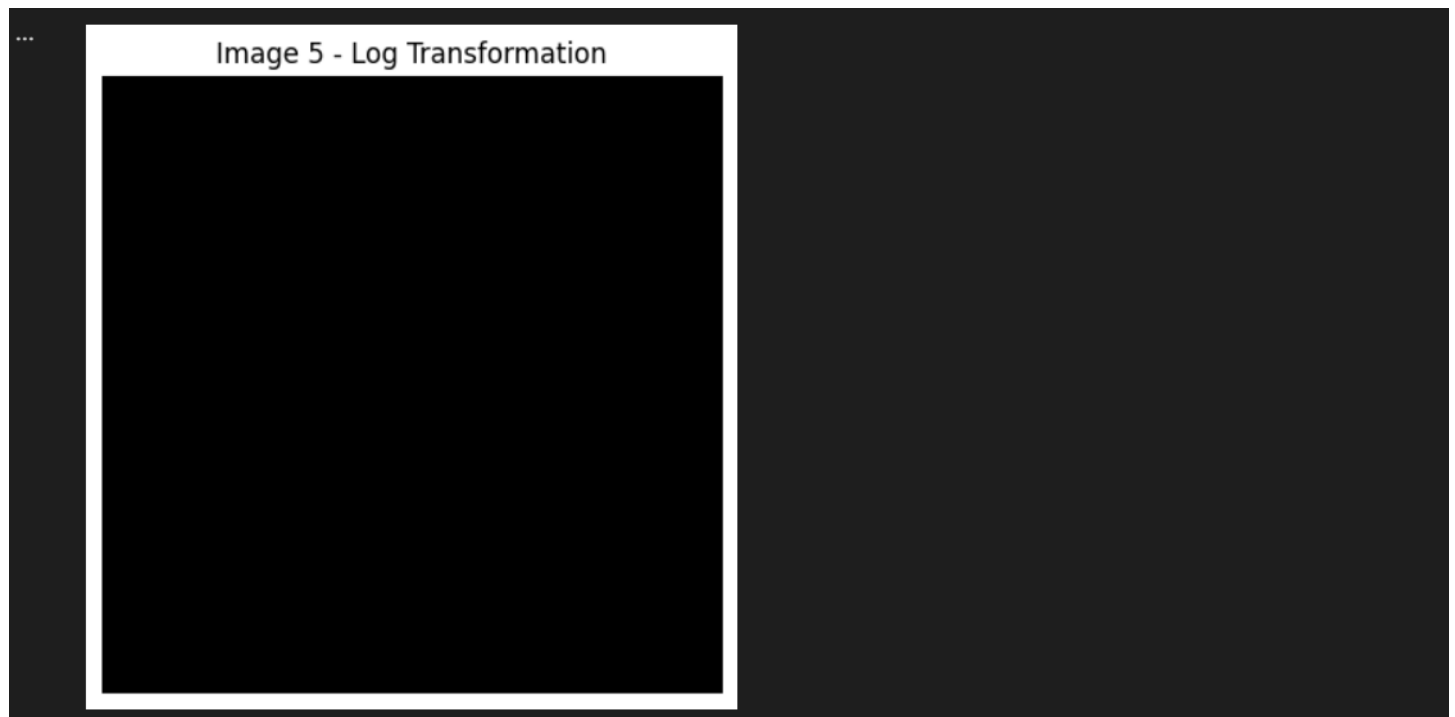
cv2.imwrite(
    os.path.join(output_folder, "image5_log.png"),
    log5
)

plt.figure(figsize=(6, 5))
plt.imshow(log5, cmap="gray")
plt.title("Image 5 - Log Transformation")
plt.axis("off")
plt.show()

```

[27] ✓ 0.0s Python

Output:



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Contrast Stretching

Code:

```

r_min = np.min(gray5)
r_max = np.max(gray5)

contrast5 = ((gray5 - r_min) / (r_max - r_min) * 255).astype(np.uint8)

cv2.imwrite(
    os.path.join(output_folder, "image5_contrast.png"),
    contrast5
)

plt.figure(figsize=(6, 5))
plt.imshow(contrast5, cmap="gray")
plt.title("Image 5 - Contrast Stretching")
plt.axis("off")
plt.show()

```

[28] ✓ 0.1s Python

Output:



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Image 5 Comparison

Code:

```

plt.figure(figsize=(12, 8))

plt.subplot(2, 2, 1)
plt.imshow(gray5, cmap="gray")
plt.title("Original Image 5")
plt.axis("off")

plt.subplot(2, 2, 2)
plt.imshow(contrast5, cmap="gray")
plt.title("Contrast Stretching")
plt.axis("off")

plt.subplot(2, 2, 3)
plt.imshow(gamma5, cmap="gray")
plt.title("Gamma Transformation")
plt.axis("off")

plt.subplot(2, 2, 4)
plt.imshow(log5, cmap="gray")
plt.title("Log Transformation")
plt.axis("off")

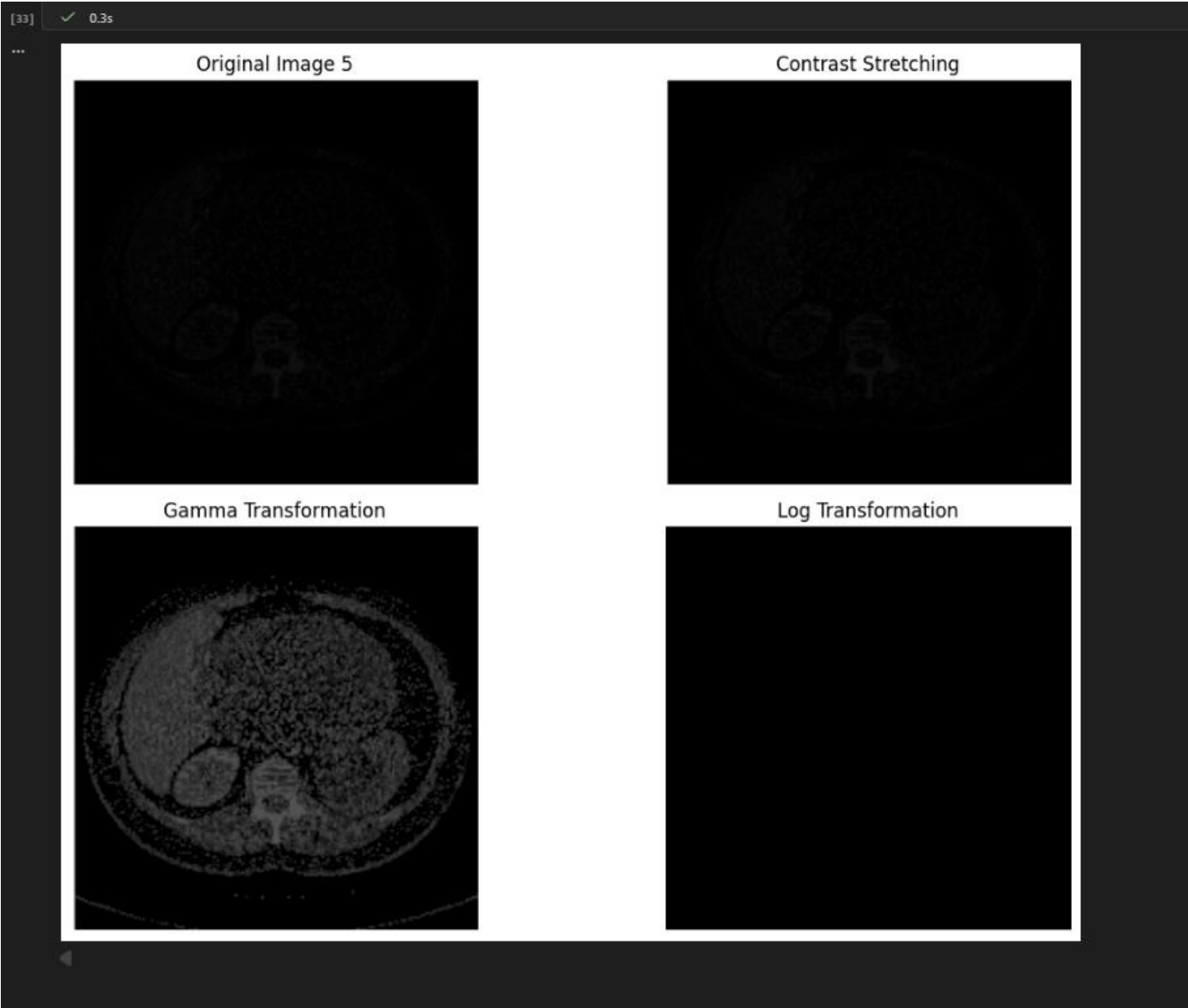
plt.tight_layout()
plt.show()

```

[33] ✓ 0.3s

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Output:



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Observation – Image 5

- The original image has very low brightness.
- Important structures are difficult to distinguish.
- Gamma transformation with increases the brightness of faint structures.
- Log transformation enhances low-intensity information.
- Contrast stretching improves the separation of different intensity levels.
- Excessive enhancement may also make noise and artifacts more visible.

Conclusion – Image 5

Gamma transformation is particularly suitable for the fifth image because the image is predominantly dark. Log transformation also improves the visibility of low-intensity structures. Contrast stretching provides additional dynamic-range enhancement.

COMMON OBSERVATIONS

The following observations can be discussed across all five images:

Contrast Stretching

- Expands the gray-level range.
- Improves contrast.
- Makes boundaries more visible.
- Can exaggerate noise in some images.

Log Transformation

- Enhances dark regions.
- Compresses bright intensities.
- Useful when low-intensity details are important.
- May reduce distinction between high-intensity regions.

Gamma Transformation

- Provides flexible brightness control.
- brightens dark images.
- darkens the image.
- Improper gamma selection can result in over-enhancement.

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Negative Transformation

- Reverses gray levels.
- May reveal details that are difficult to see in the original image.
- Does not increase the actual information content.

LIMITATIONS

Gray-level transformations have several limitations:

1. They modify pixel intensities but do not create new information.
2. They cannot completely recover details lost during image acquisition.
3. Enhancement may also amplify noise.
4. Choosing unsuitable parameters can produce over-enhancement.
5. One transformation may work well for one image but poorly for another.
6. Gray-level transformations alone are not sufficient for all types of degradation.
7. Images with severe noise may require spatial filtering.
8. Images with blur may require restoration/deconvolution techniques.
9. Images with complex illumination problems may require more advanced methods.

The assignment specifically asks the final conclusion to address both the **effectiveness and limitations** of the gray-level techniques.

APPLICATIONS

Medical Imaging:

Gray-level enhancement can improve visibility of structures in X-rays, CT scans, MRI images, and other medical images.

Satellite Image Processing:

Low-contrast satellite images can be enhanced to improve the visibility of terrain and environmental features.

Industrial Inspection:

Enhancement can help identify defects, cracks, and irregularities in manufactured components.

Photography:

Gamma and contrast transformations can improve brightness and contrast in photographs.

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Document Processing:

Gray-level enhancement can improve the visibility of text and characters before OCR.

Security and Surveillance:

Dark surveillance images can be enhanced to make objects and structures more visible.

Microscopic Imaging:

Low-contrast microscopic images can be enhanced to reveal cellular structures.

Remote Sensing:

Intensity transformations can improve interpretation of images acquired under different illumination conditions

Final Conclusion

The experiment successfully demonstrated the use of different gray-level transformation techniques for enhancing five degraded images. The images provided in the assignment exhibited different visual characteristics, including low contrast, dark intensity levels, and limited visibility of important structures. Contrast stretching was useful for images having a limited range of gray levels because it expanded the available intensity range. Logarithmic transformation enhanced information in darker regions by expanding low-intensity values. Power-law/gamma transformation provided flexible brightness control and was particularly useful for very dark images when $\gamma < 1$.

The experiment also demonstrated that there is no single gray-level transformation that is optimal for every degraded image. The selection of the transformation and its parameters depends on the characteristics of the input image.

The techniques improved the visual appearance of the degraded images, but they also have limitations. They cannot recover information that was completely lost during image acquisition and may sometimes amplify noise or unwanted variations. Therefore, for severely degraded images, gray-level transformations may need to be combined with other techniques such as spatial filtering, noise reduction, histogram-based methods, or image restoration.

Overall, the experiment provided practical understanding of how gray-level transformations can be selected and applied to improve the visual quality and interpretability of degraded digital images.

GitHub:

<https://github.com/Devaharshagoud/DSIP-Experiments.git>